

LLMs as few-shot learners

NLP: Fall 2024

Anoop Sarkar

"Language provides a natural domain for the study of artificial intelligence, as the vast majority of reasoning tasks can be efficiently expressed and evaluated in language, and the world's text provides a wealth of data for unsupervised learning via generative modeling."

- OpenAI

Improving Language Understanding by Generative Pre-Training

GPT1

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GPT1

Pre-training an autoregressive language model

BooksCorpus: 7K
unpublished books
(1B words)

- Start with a large amount of unlabeled data $\mathcal{U} = \{u_1, \dots, u_n\}$
- Pre-training objective: Maximize the likelihood of predicting the next token

$$L_i(\mathcal{U}) = \sum_i \log P(u_i \mid u_{i-k}, \dots, u_{i-1}; \Theta)$$

$U = (u_{-k}, \dots, u_{-1})$ is the context vector of tokens

- This is equivalent to training a Transformer decoder

n is the number of Transformer layers

$$h_0 = U \boxed{W_e} + W_p$$

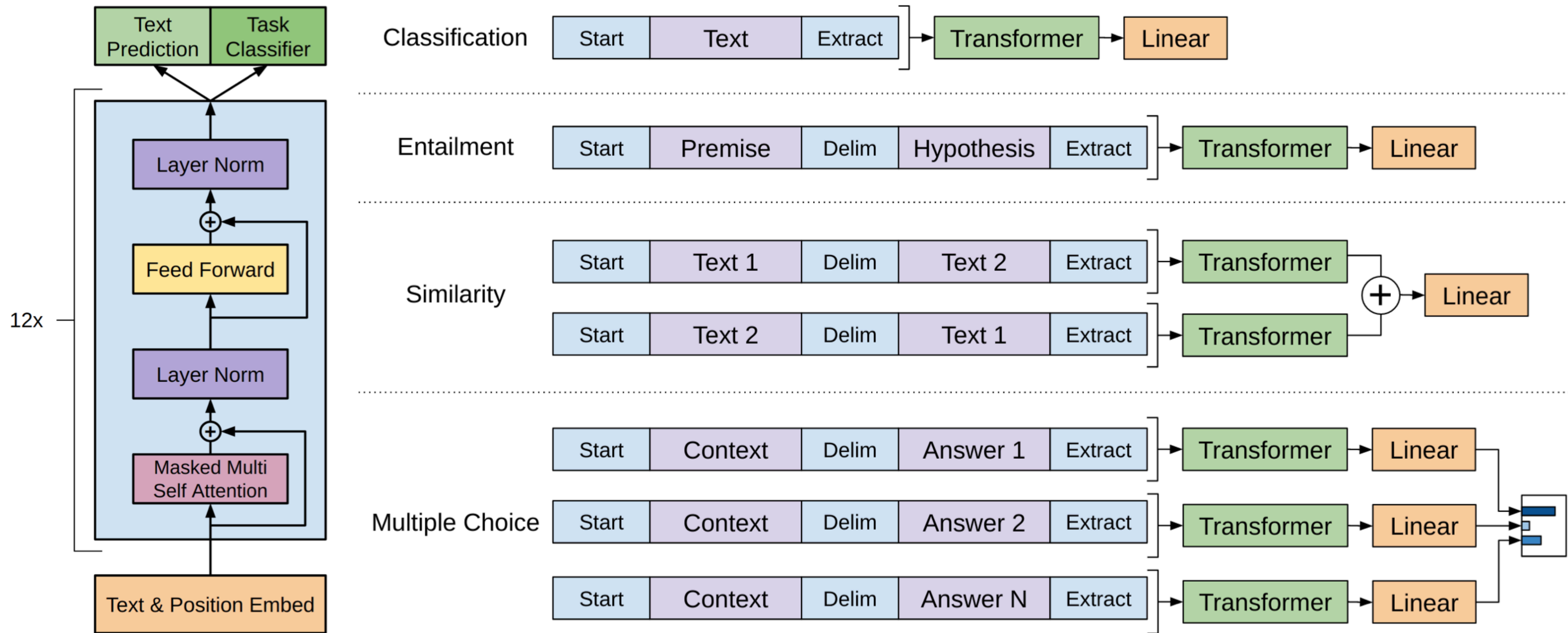
W_e is the token embedding matrix

$$h_\ell = \text{transformer_block}(h_{\ell-1}) \forall \ell \in [1, n]$$

W_p is the position embedding matrix

$$P(u) = \text{softmax}(h_n \boxed{W_e^T})$$

- Directionality is needed to generate a well-formed probability distribution



This setup was for fine-tuning GPT1 but also works for in-context learning in GPT2 and GPT3.

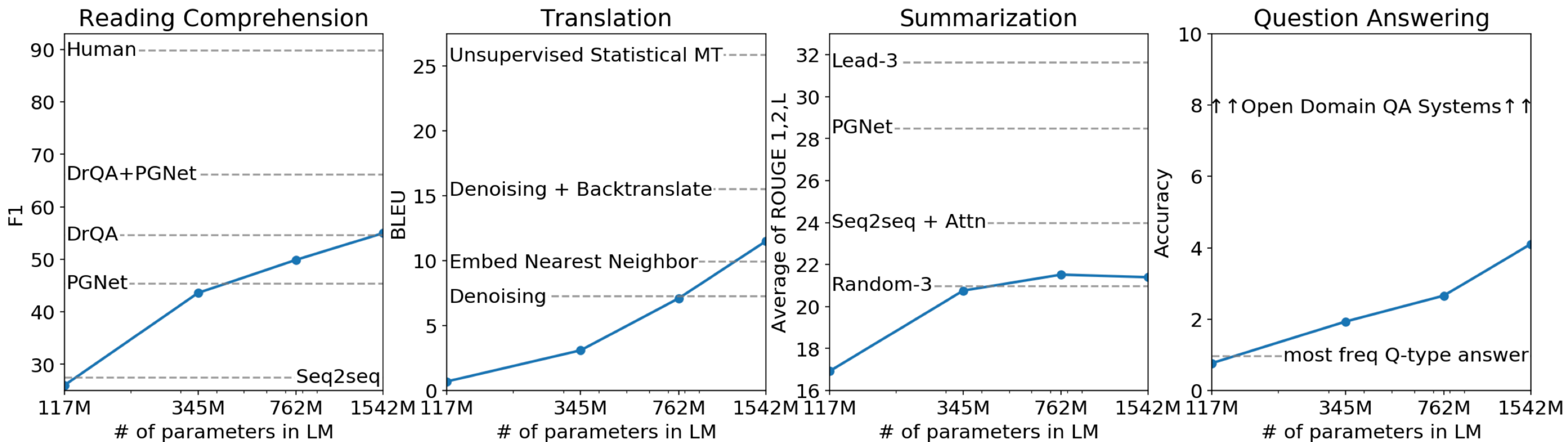
The GPT2 paper

Language Models are Unsupervised Multitask Learners

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[https://cdn.openai.com/better-language-models/
language_models_are_unsupervised_multitask_learners.pdf](https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf)

Feb 2019



WebText corpus

- Train on web scale corpus but with more reliable data compared to the CommonCrawl.
- English-only, so language detection is used
- Outgoing links from reddit (with at least 3 karma)
- No reddit data was used, instead use the content of the web sites linked on reddit discussions
- 8M documents with 40GB of text

Language detection: <https://github.com/CLD2Owners/cld2>

News site scraping: <https://github.com/codelucas/newspaper>

”I’m not the cleverest man in the world, but like they say in French: **Je ne suis pas un imbécile** [I’m not a fool].

In a now-deleted post from Aug. 16, Soheil Eid, Tory candidate in the riding of Joliette, wrote in French: **”Mentez mentez, il en restera toujours quelque chose,”** which translates as, **”Lie lie and something will always remain.”**

“I hate the word ‘perfume,’” Burr says. ‘It’s somewhat better in French: ‘**parfum.**’

If listened carefully at 29:55, a conversation can be heard between two guys in French: **“-Comment on fait pour aller de l’autre côté? -Quel autre côté?”**, which means **“- How do you get to the other side? - What side?”**.

If this sounds like a bit of a stretch, consider this question in French: **As-tu aller au cinéma?**, or **Did you go to the movies?**, which literally translates as Have-you to go to movies/theater?

“Brevet Sans Garantie Du Gouvernement”, translated to English: **“Patented without government warranty”**.

Table 1. Examples of naturally occurring demonstrations of English to French and French to English translation found throughout the WebText training set.

Parameters	Layers	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542M	48	1600

Table 2. Architecture hyperparameters for the 4 model sizes.

Perplexity Results

	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText2 (PPL)	PTB (PPL)	enwik8 (BPB)	text8 (BPC)	WikiText103 (PPL)	1BW (PPL)
SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	1.08	18.3	21.8
117M	35.13	45.99	87.65	83.4	29.41	65.85	1.16	1.17	37.50	75.20
345M	15.60	55.48	92.35	87.1	22.76	47.33	1.01	1.06	26.37	55.72
762M	10.87	60.12	93.45	88.0	19.93	40.31	0.97	1.02	22.05	44.575
1542M	8.63	63.24	93.30	89.05	18.34	35.76	0.93	0.98	17.48	42.16

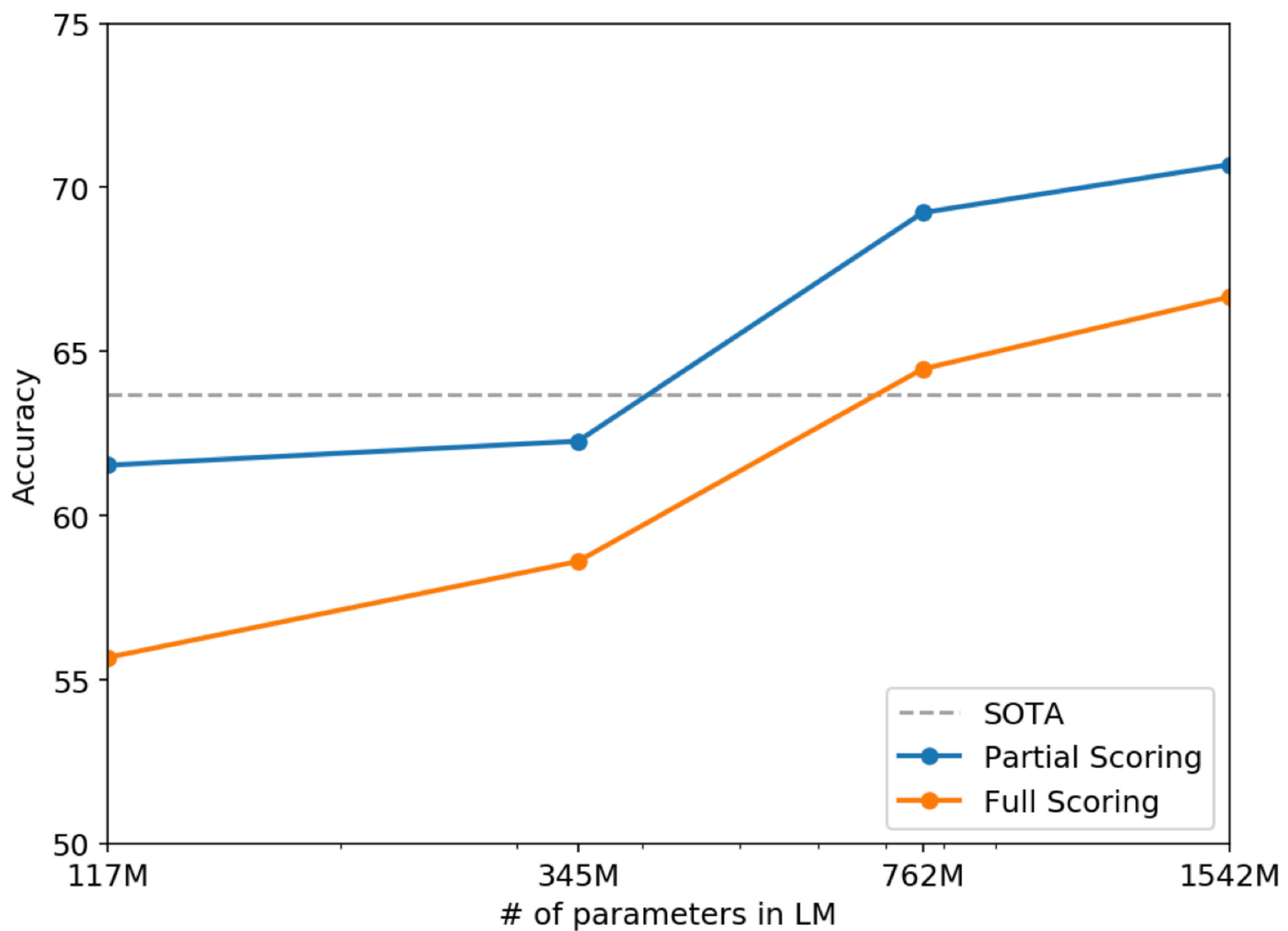


Figure 3. Performance on the Winograd Schema Challenge as a function of model capacity.

Language Models are Few-Shot Learners

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Tom Henighan

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Jeffrey Wu

Clemens Winter

Christopher Hesse

Mark Chen

Eric Sigler

Mateusz Litwin

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Benjamin Chess

Jack Clark

Christopher Berner

Sam McCandlish

Alec Radford

Ilya Sutskever

Dario Amodei

Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



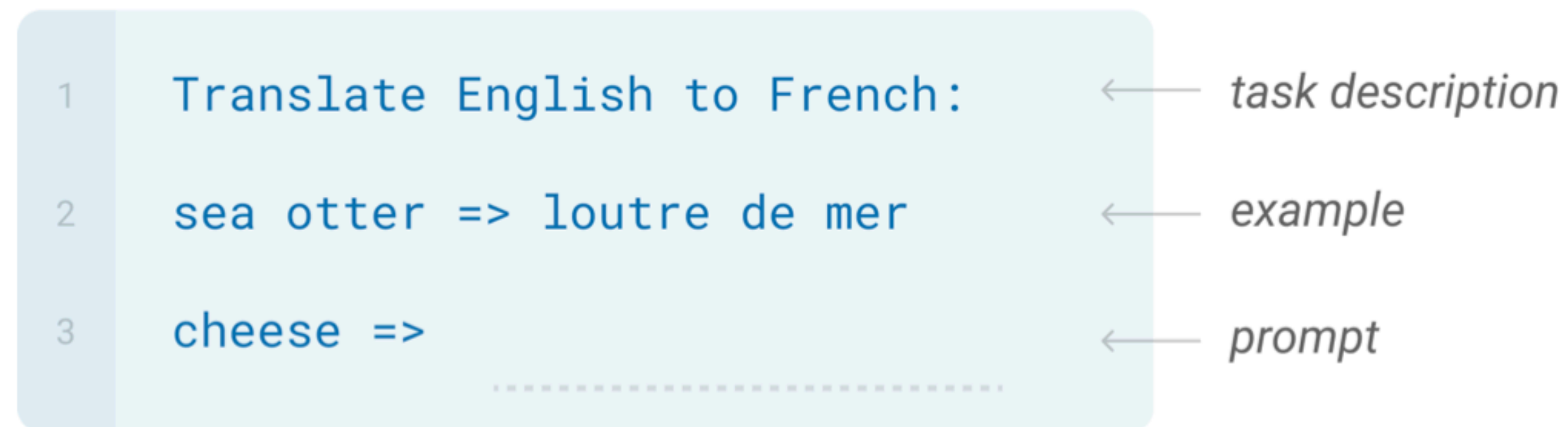
Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.



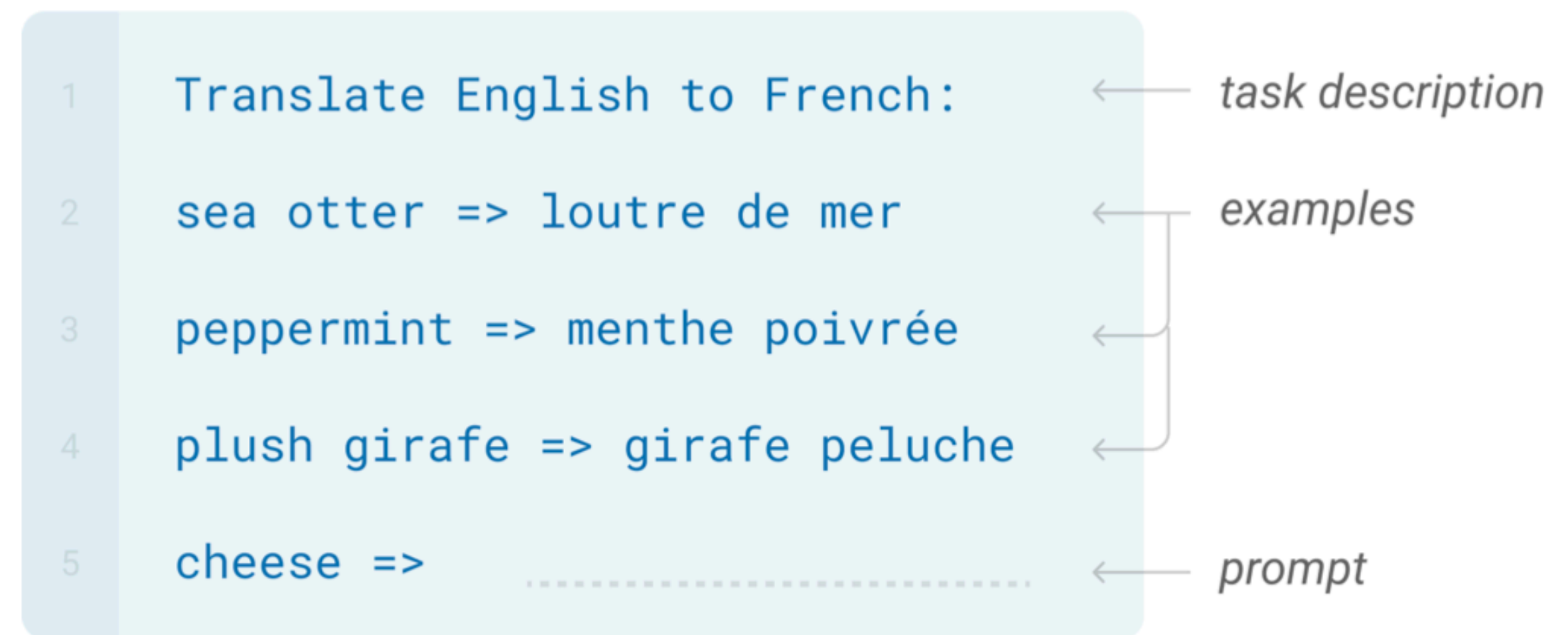
One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.



Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



Fine-tuning fails at scale

- LLMs $>10\text{B}$ parameters are very difficult to fine-tune and requires a big compute budget
- So in-context learning using a long prompt or prefix is needed to coax the answer from a "predict the next token" approach to solving multiple tasks
- Pre-training on web-scale text can observe many different tasks in-context during training in the inner loop (per batch)
- Gradient descent improves the model representations based on next token prediction over many batch updates in the outer loop

outer loop

Learning via SGD during unsupervised pre-training

inner loop

In-context learning

↑
sequence #1

1	$5 + 8 = 13$
2	$7 + 2 = 9$
3	$1 + 0 = 1$
4	$3 + 4 = 7$
5	$5 + 9 = 14$
6	$9 + 8 = 17$

In-context learning

↑
sequence #2

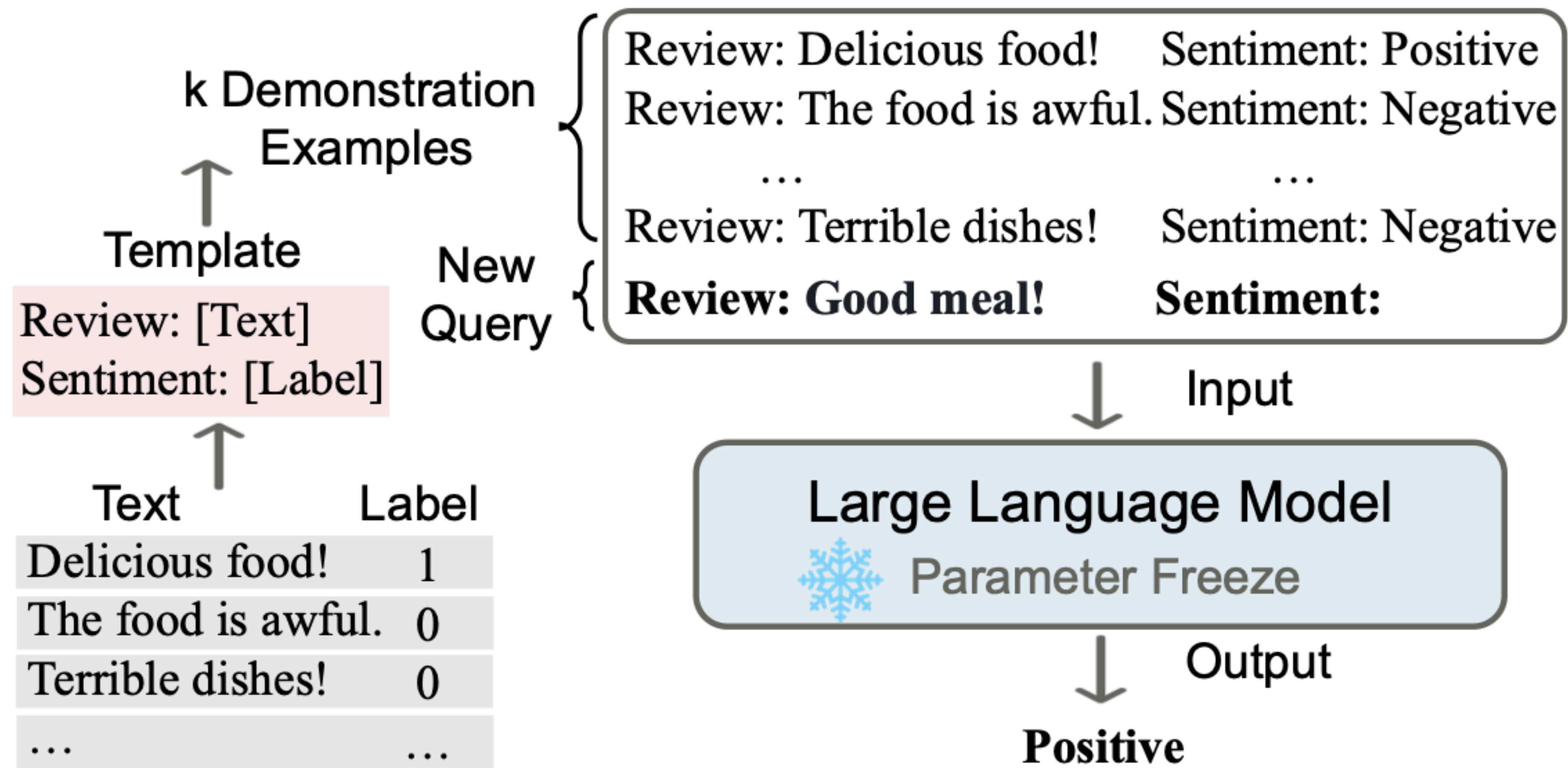
1	gaot => goat
2	sakne => snake
3	brid => bird
4	fsih => fish
5	dcuk => duck
6	cmihp => chimp

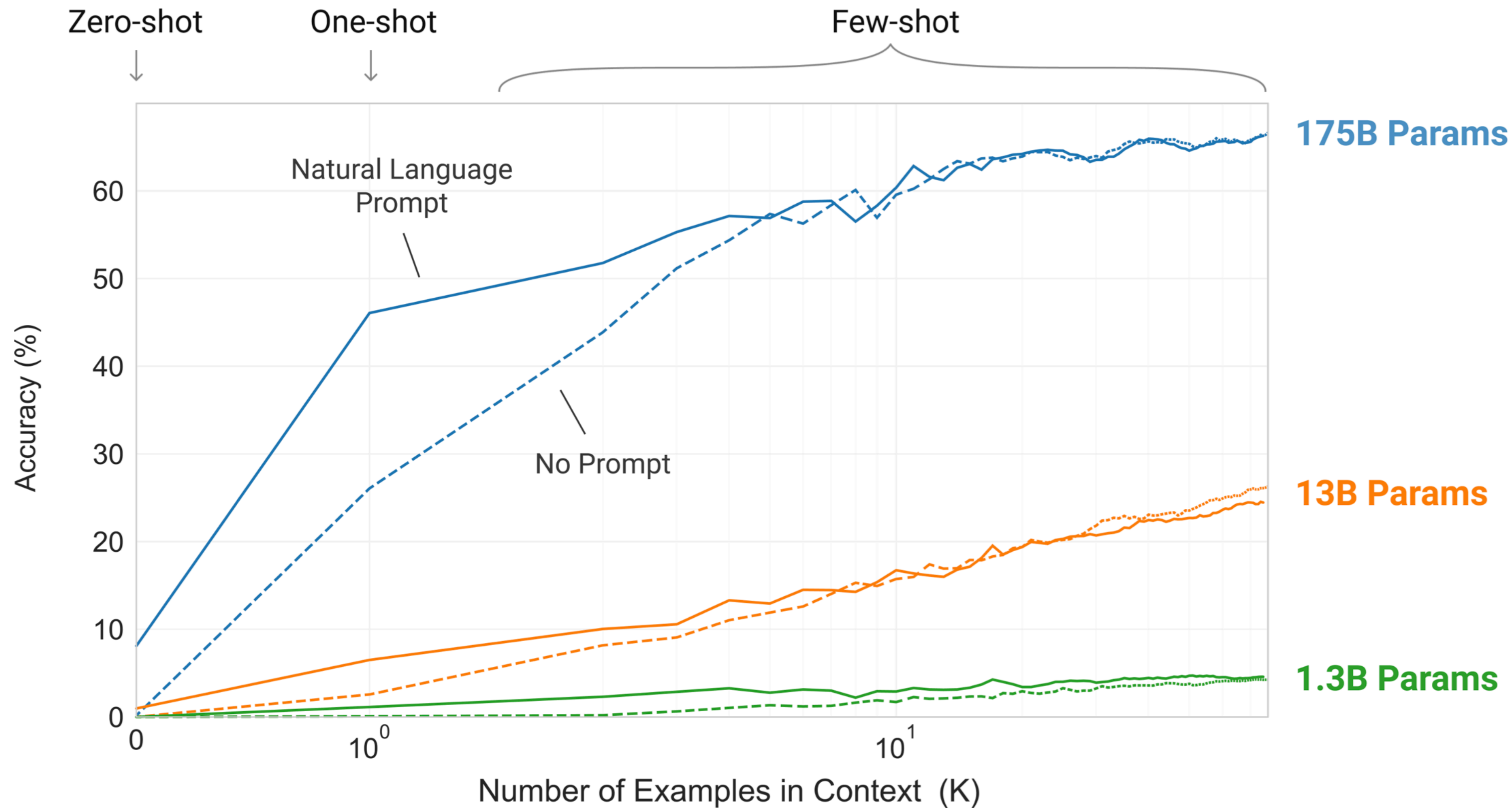
In-context learning

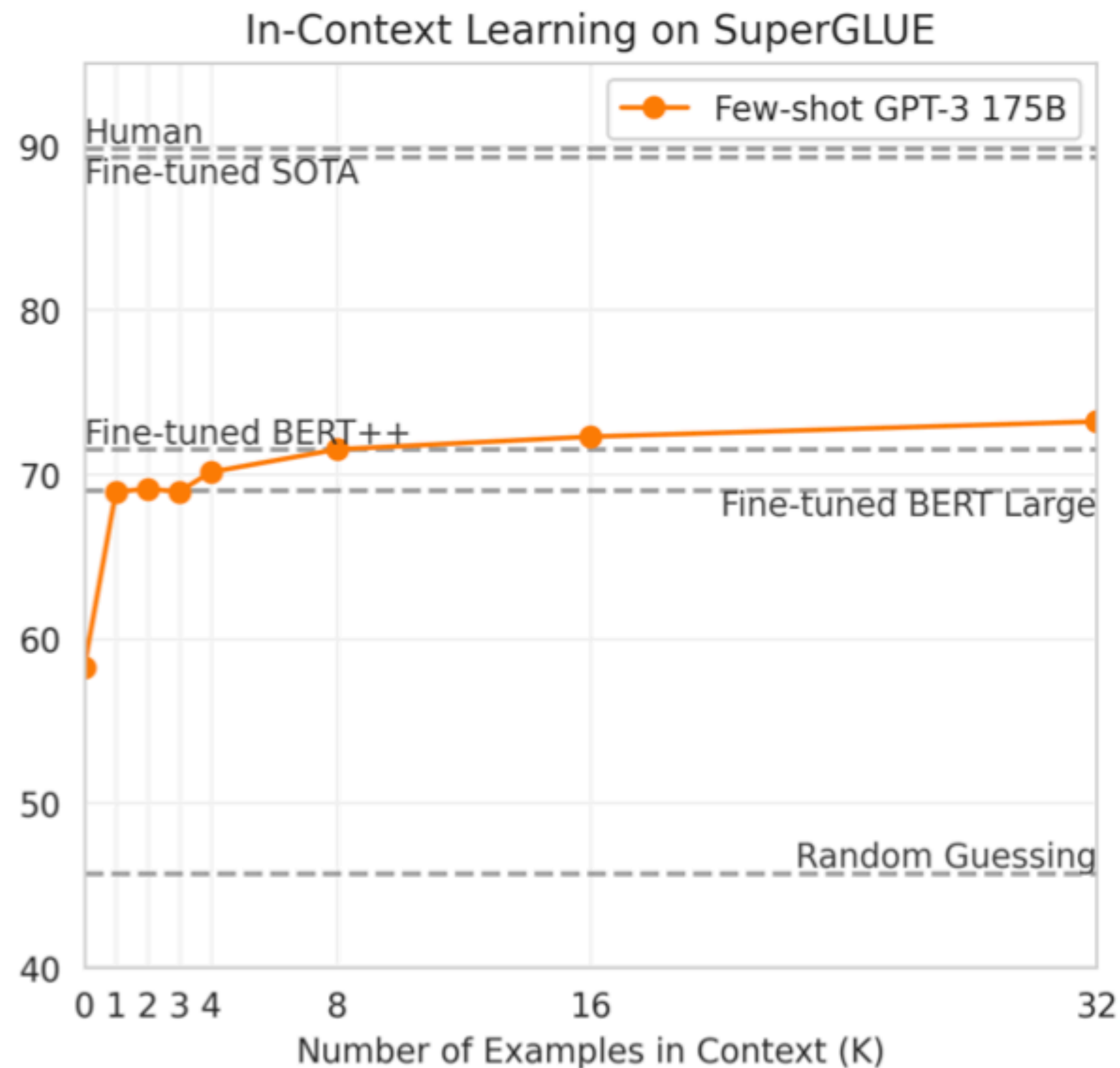
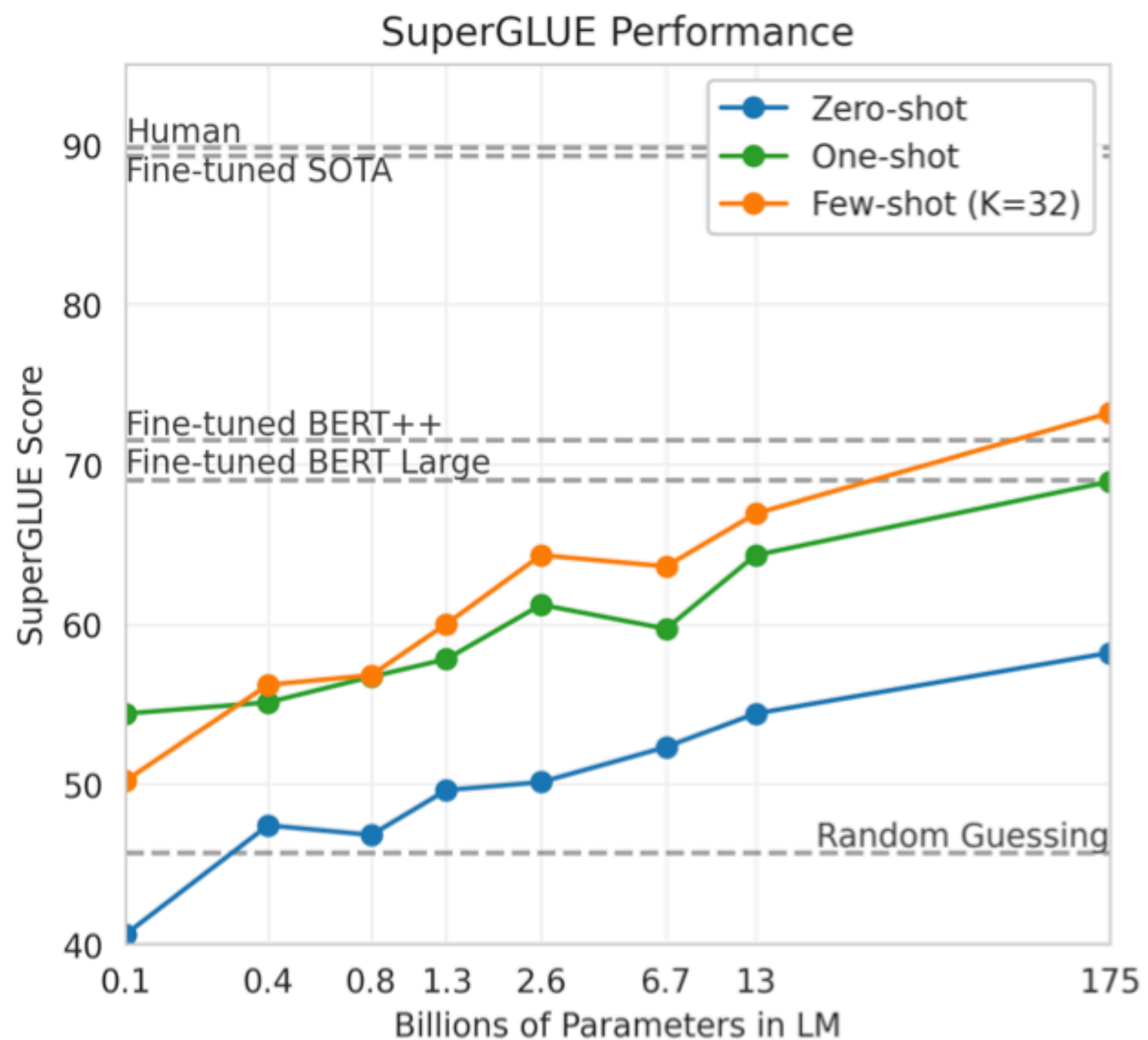
↑
sequence #3

1	thanks => merci
2	hello => bonjour
3	mint => menthe
4	wall => mur
5	otter => loutre
6	bread => pain

Inference time







Performance on SuperGLUE increases with number of examples in context. We find the difference in performance between the BERT-Large and BERT++ to be roughly equivalent to the difference between GPT-3 with one example per context versus eight examples per context.

	SuperGLUE Average	BoolQ Accuracy	CB Accuracy	CB F1	COPA Accuracy	RTE Accuracy
Fine-tuned SOTA	89.0	91.0	96.9	93.9	94.8	92.5
Fine-tuned BERT-Large	69.0	77.4	83.6	75.7	70.6	71.7
GPT-3 Few-Shot	71.8	76.4	75.6	52.0	92.0	69.0
	WiC Accuracy	WSC Accuracy	MultiRC Accuracy	MultiRC F1a	ReCoRD Accuracy	ReCoRD F1
Fine-tuned SOTA	76.1	93.8	62.3	88.2	92.5	93.3
Fine-tuned BERT-Large	69.6	64.6	24.1	70.0	71.3	72.0
GPT-3 Few-Shot	49.4	80.1	30.5	75.4	90.2	91.1

Table 3.5: Performance of GPT-3 on SuperGLUE compared to fine-tuned baselines and SOTA. All results are reported on the test set. GPT-3 few-shot is given a total of 32 examples within the context of each task and performs no gradient updates.

Setting	LAMBADA (acc)	LAMBADA (ppl)	StoryCloze (acc)	HellaSwag (acc)
SOTA	68.0 ^a	8.63 ^b	91.8^c	85.6^d
GPT-3 Zero-Shot	76.2	3.00	83.2	78.9
GPT-3 One-Shot	72.5	3.35	84.7	78.1
GPT-3 Few-Shot	86.4	1.92	87.7	79.3

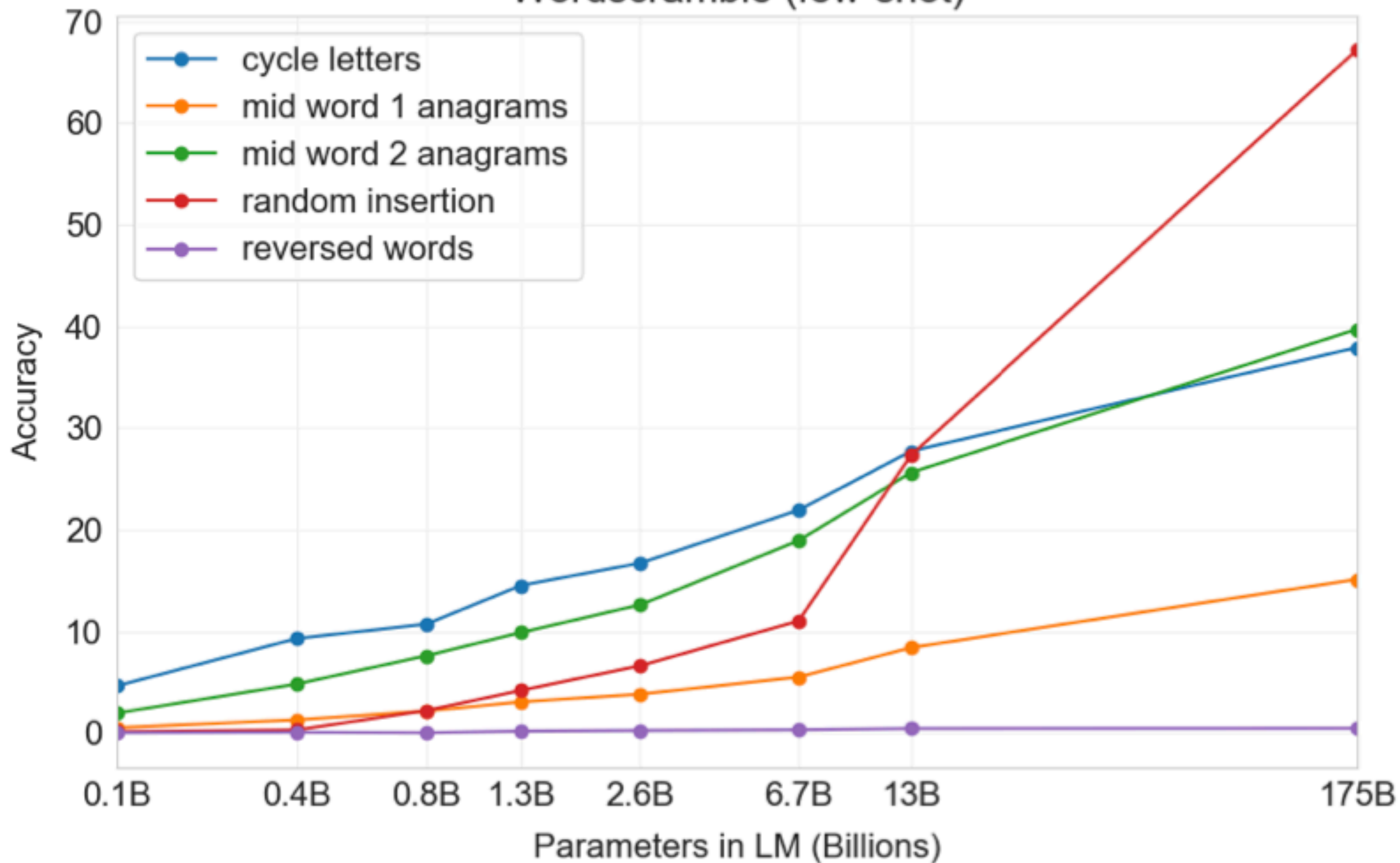
Setting	NaturalQS	WebQS	TriviaQA
RAG (Fine-tuned, Open-Domain) [LPP⁺20]	44.5	45.5	68.0
T5-11B+SSM (Fine-tuned, Closed-Book) [RRS20]	36.6	44.7	60.5
T5-11B (Fine-tuned, Closed-Book)	34.5	37.4	50.1
GPT-3 Zero-Shot	14.6	14.4	64.3
GPT-3 One-Shot	23.0	25.3	68.0
GPT-3 Few-Shot	29.9	41.5	71.2

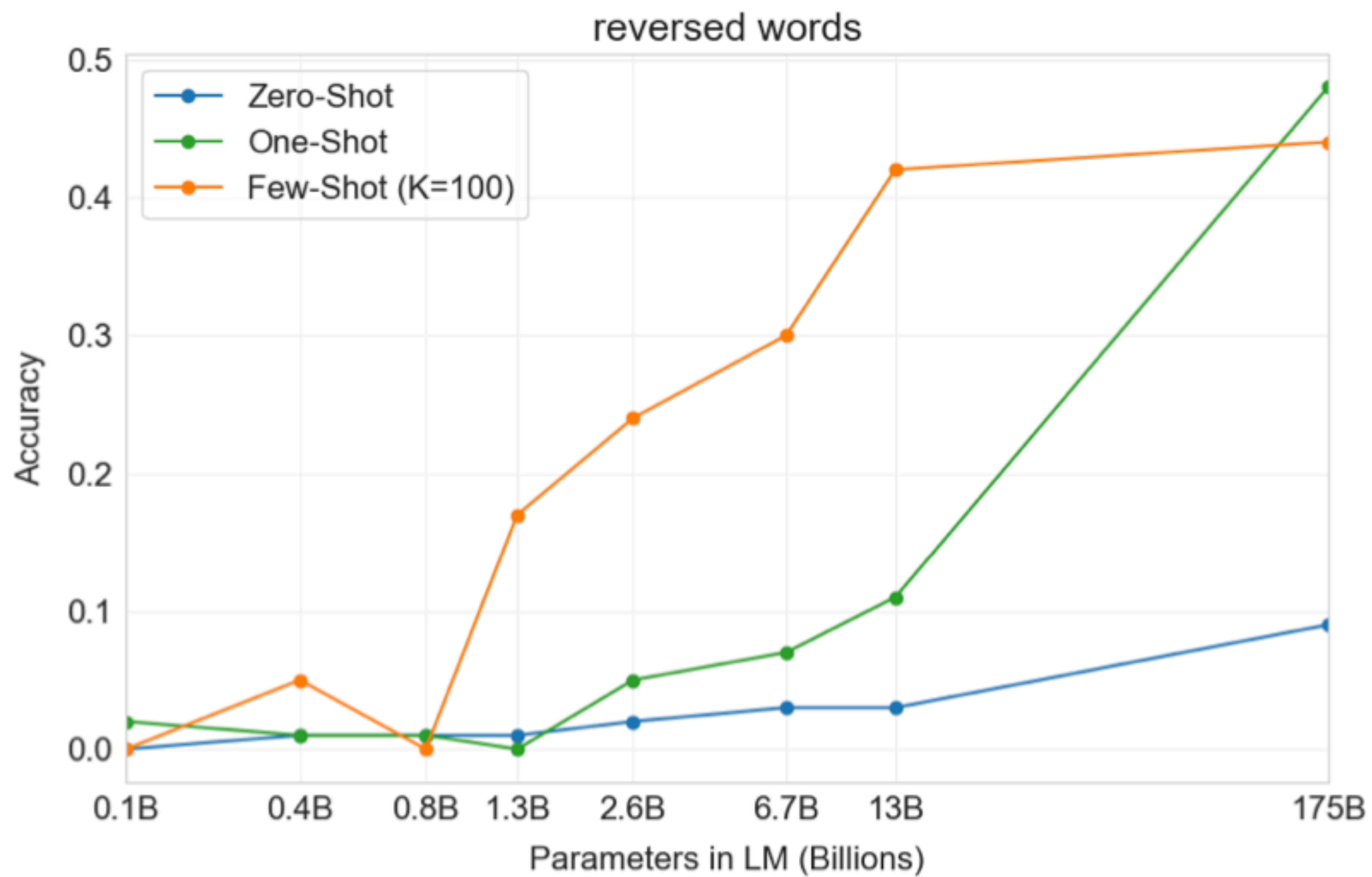
Setting	ARC (Easy)	ARC (Challenge)	CoQA	DROP
Fine-tuned SOTA	92.0^a	78.5^b	90.7^c	89.1^d
GPT-3 Zero-Shot	68.8	51.4	81.5	23.6
GPT-3 One-Shot	71.2	53.2	84.0	34.3
GPT-3 Few-Shot	70.1	51.5	85.0	36.5

WMT 2014

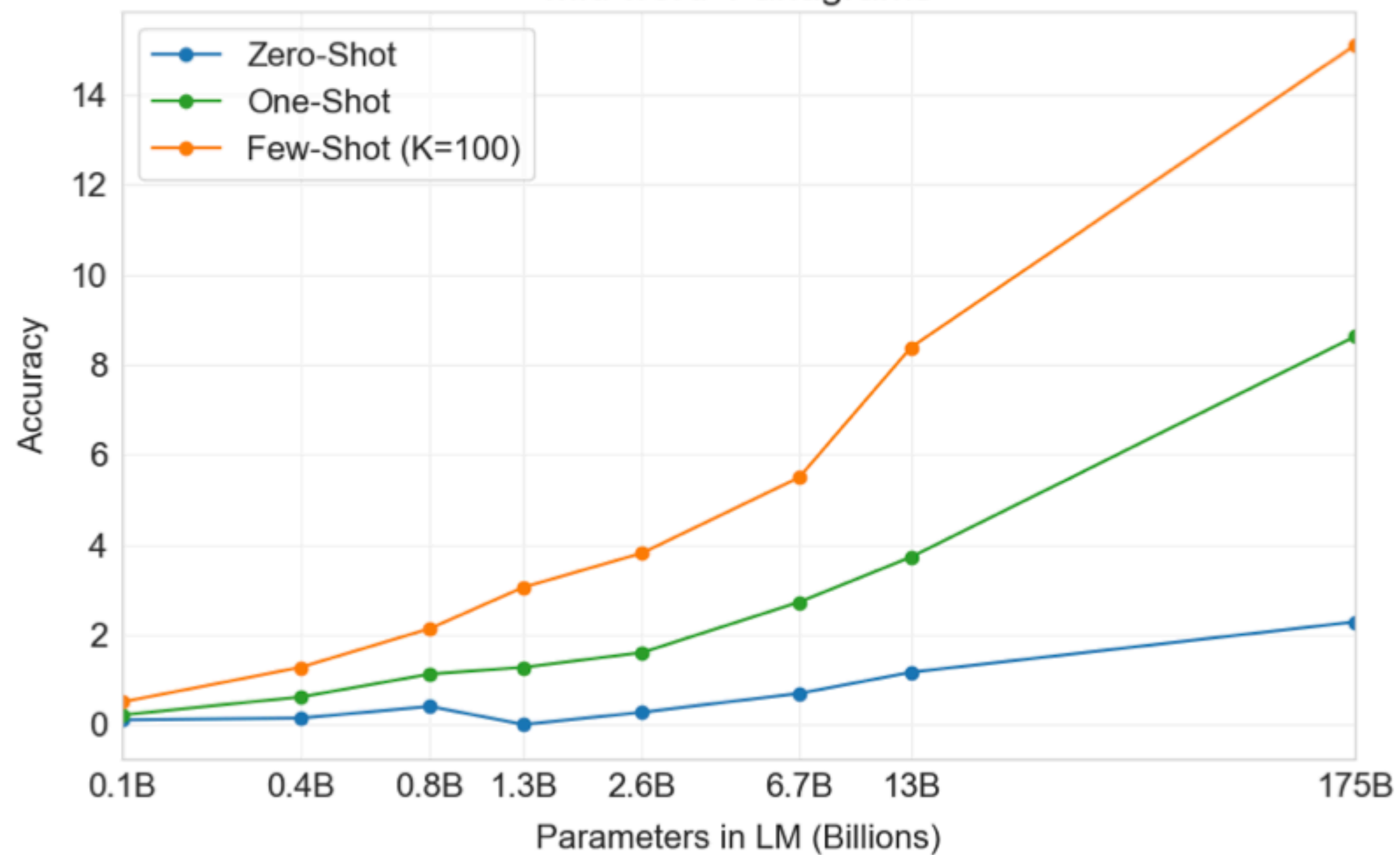
Setting	En→Fr	Fr→En	En→De	De→En	En→Ro	Ro→En
SOTA (Supervised)	45.6^a	35.0 ^b	41.2^c	40.2 ^d	38.5^e	39.9^e
XLM [LC19]	33.4	33.3	26.4	34.3	33.3	31.8
MASS [STQ ⁺ 19]	<u>37.5</u>	34.9	28.3	35.2	<u>35.2</u>	33.1
mBART [LGG ⁺ 20]	-	-	<u>29.8</u>	34.0	35.0	30.5
GPT-3 Zero-Shot	25.2	21.2	24.6	27.2	14.1	19.9
GPT-3 One-Shot	28.3	33.7	26.2	30.4	20.6	38.6
GPT-3 Few-Shot	32.6	<u>39.2</u>	29.7	<u>40.6</u>	21.0	<u>39.5</u>

Wordscramble (few-shot)

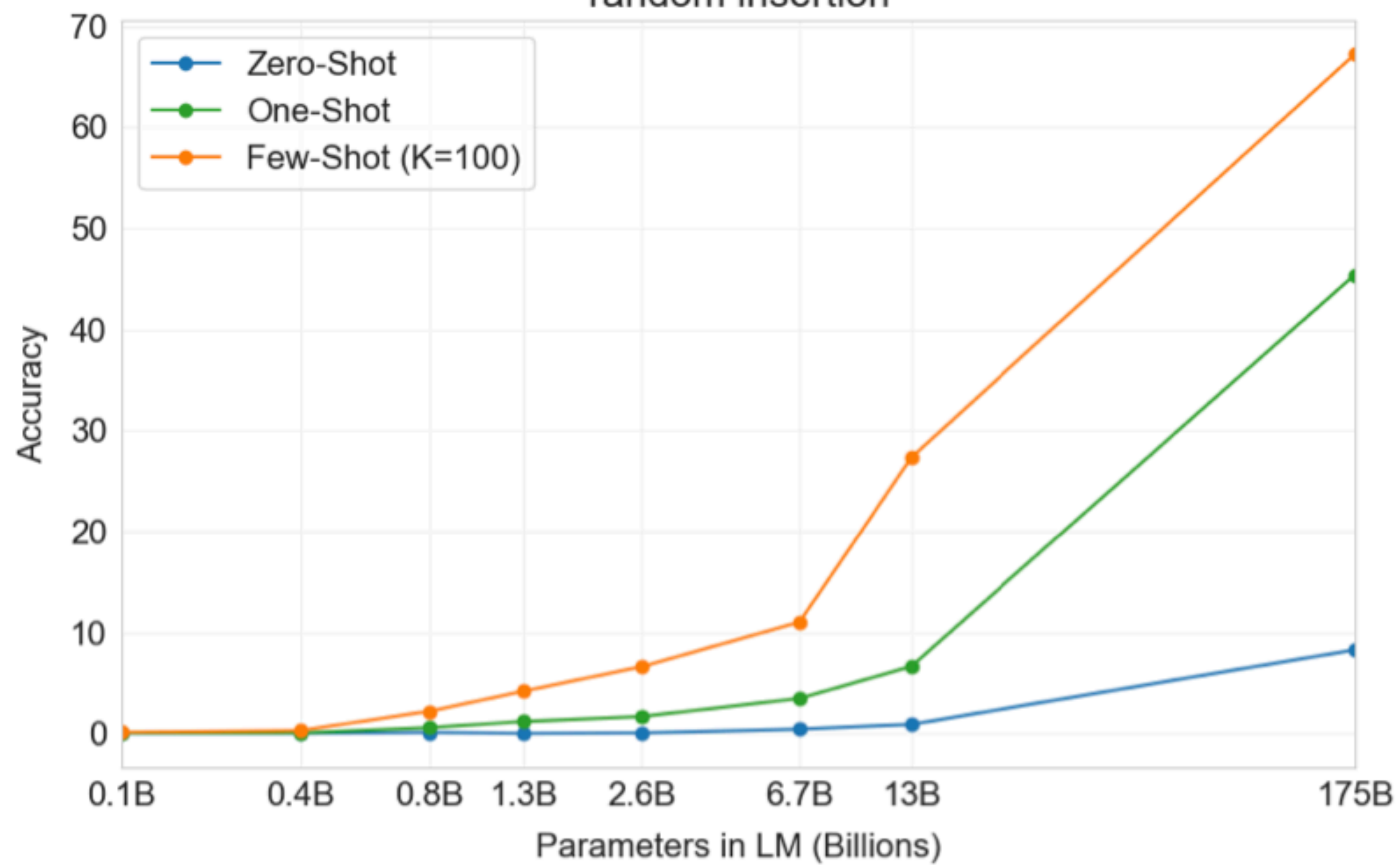




mid word 1 anagrams



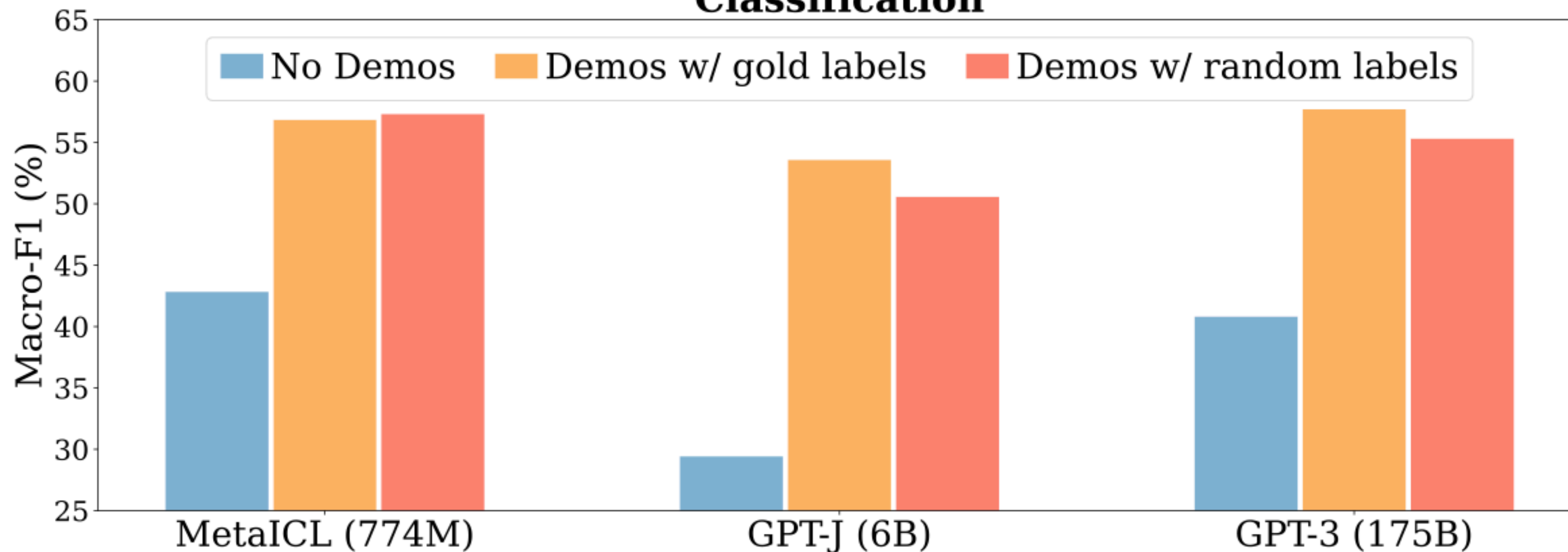
random insertion



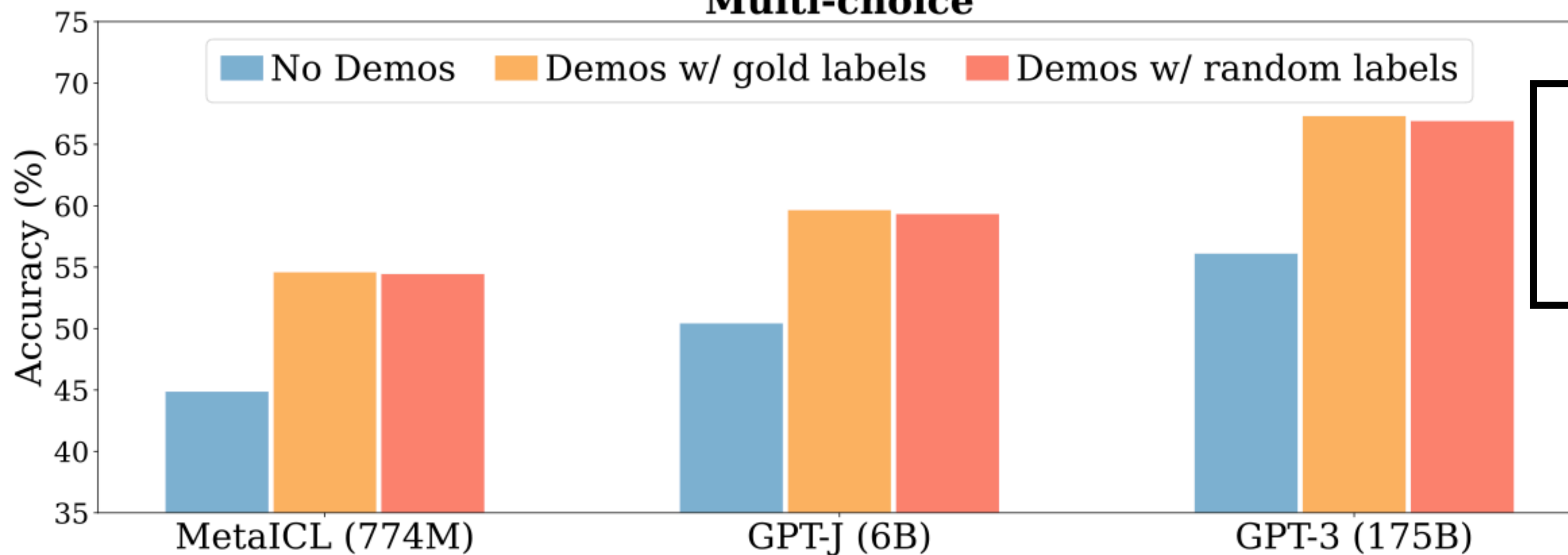
Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?

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Mike Lewis² Hannaneh Hajishirzi^{1,3} Luke Zettlemoyer^{1,2}
¹University of Washington ²Meta AI ³Allen Institute for AI

Classification



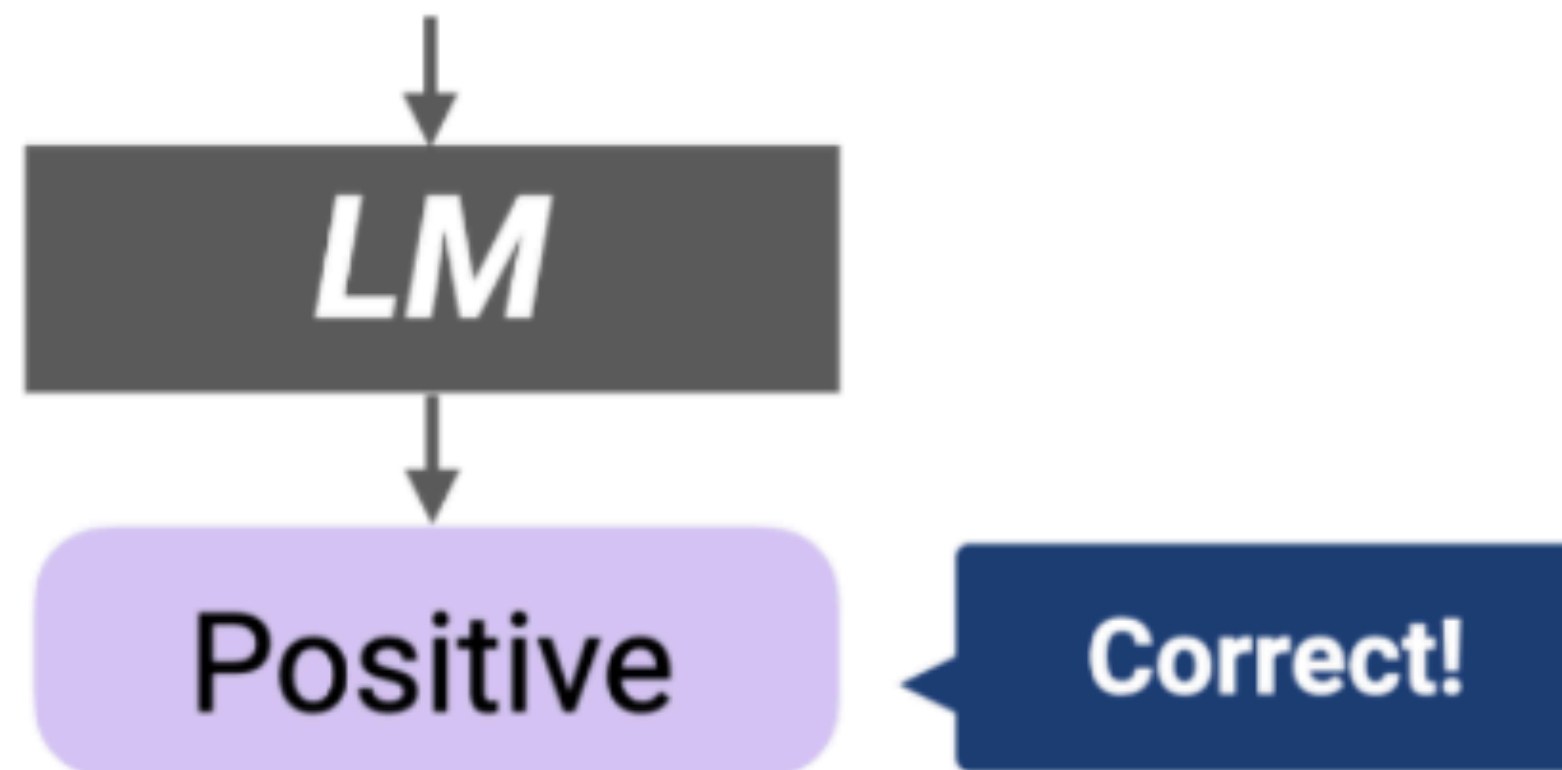
Multi-choice



ground truth
labels don't
matter!

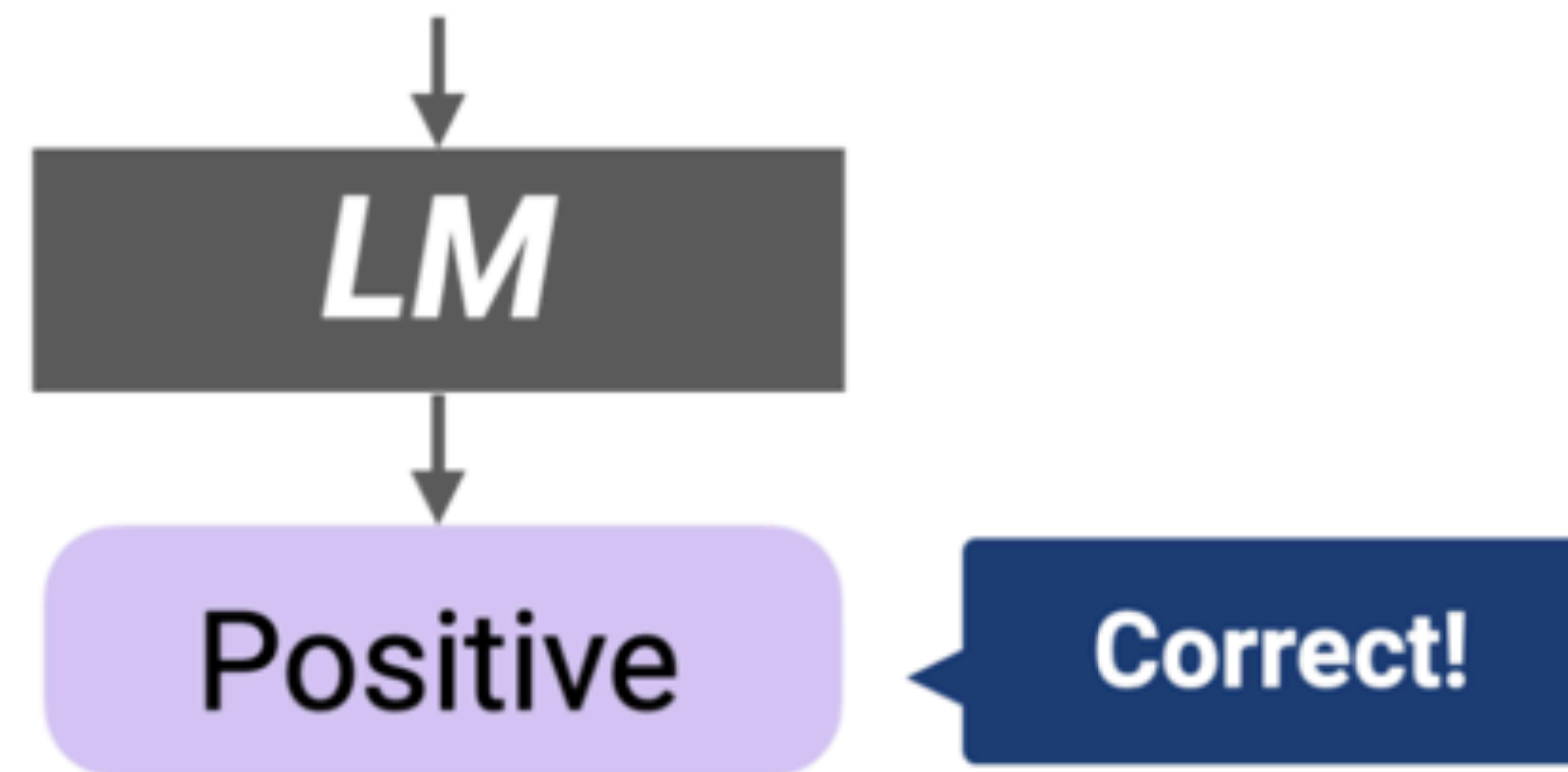
ground truth
labels

Circulation revenue has increased by 5% in Finland.	\n	Positive
Panostaja did not disclose the purchase price.	\n	Neutral
Paying off the national debt will be extremely painful.	\n	Negative
The company anticipated its operating profit to improve.	\n	_____



replace true labels with
random labels

Circulation revenue has increased by 5% in Finland. \n **Neutral**
Panostaja did not disclose the purchase price. \n **Negative**
Paying off the national debt will be extremely painful. \n **Positive**
The company anticipated its operating profit to improve. \n _____



Why does in-context learning work?

Four hypotheses

1. The input-label mapping, whether each input x_i is paired with the correct label y_i (not true)
2. The distribution that the input $x_1, \dots, x_k$ are from (is it from a sports article, or business news?)
3. The output label space $y_1, \dots, y_k$
4. The format of the demonstration, e.g. $x \quad // \quad y$; Input: x Output: y ; etc.

Demonstrations

Distribution of inputs

Label space

***Format
(The use
of pairs)***

Circulation revenue has increased by 5% in Finland.

\n

Positive

Panostaja did not disclose the purchase price.

\n

Neutral

Paying off the national debt will be extremely painful.

\n

Negative

Test example

Input-label mapping

The acquisition will have an immediate positive impact. \n

?

Colour-printed lithograph. Very good condition.

\n Neutral

Many accompanying marketing ... meaning.

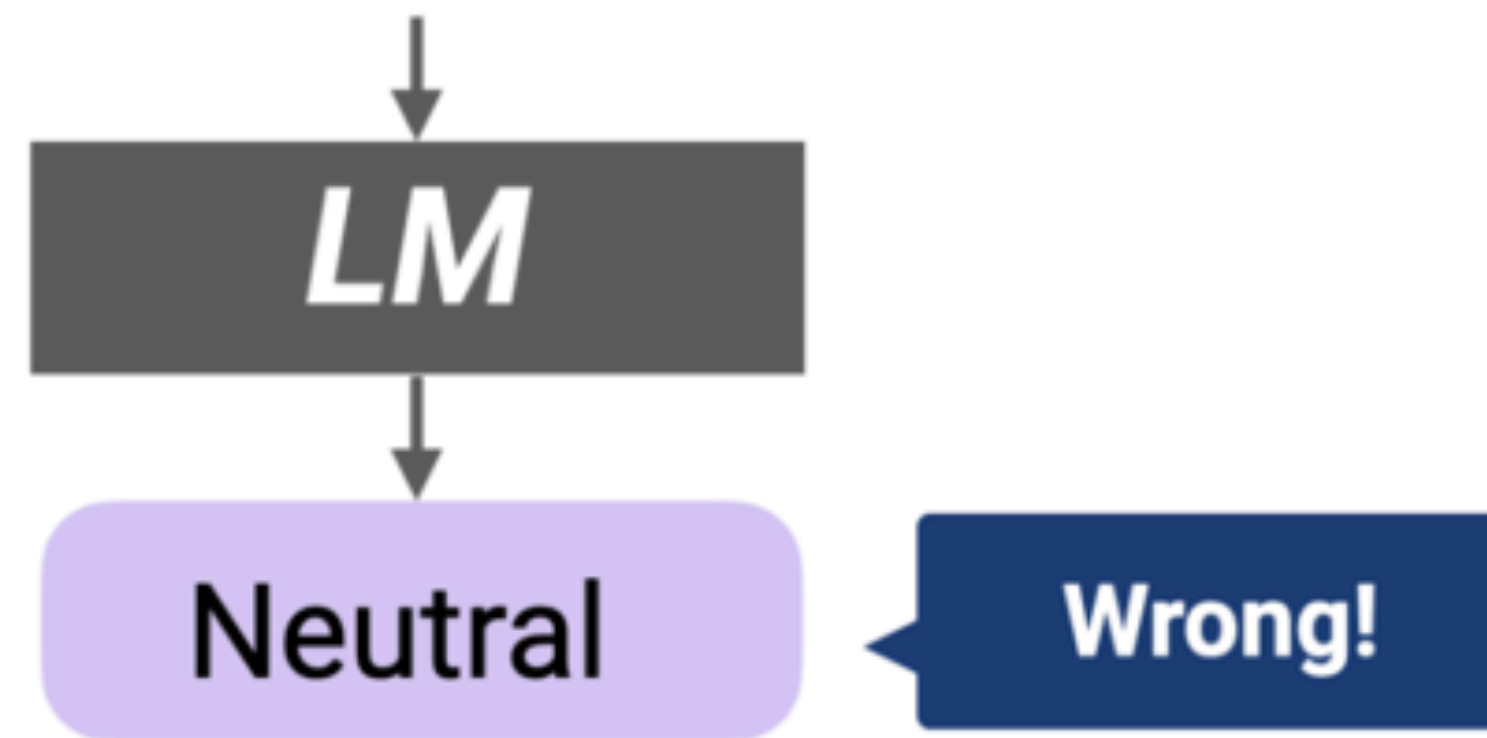
\n Negative

In case you are interested in learning more about ...

\n Positive

The company anticipated its operating profit to improve. \n _____

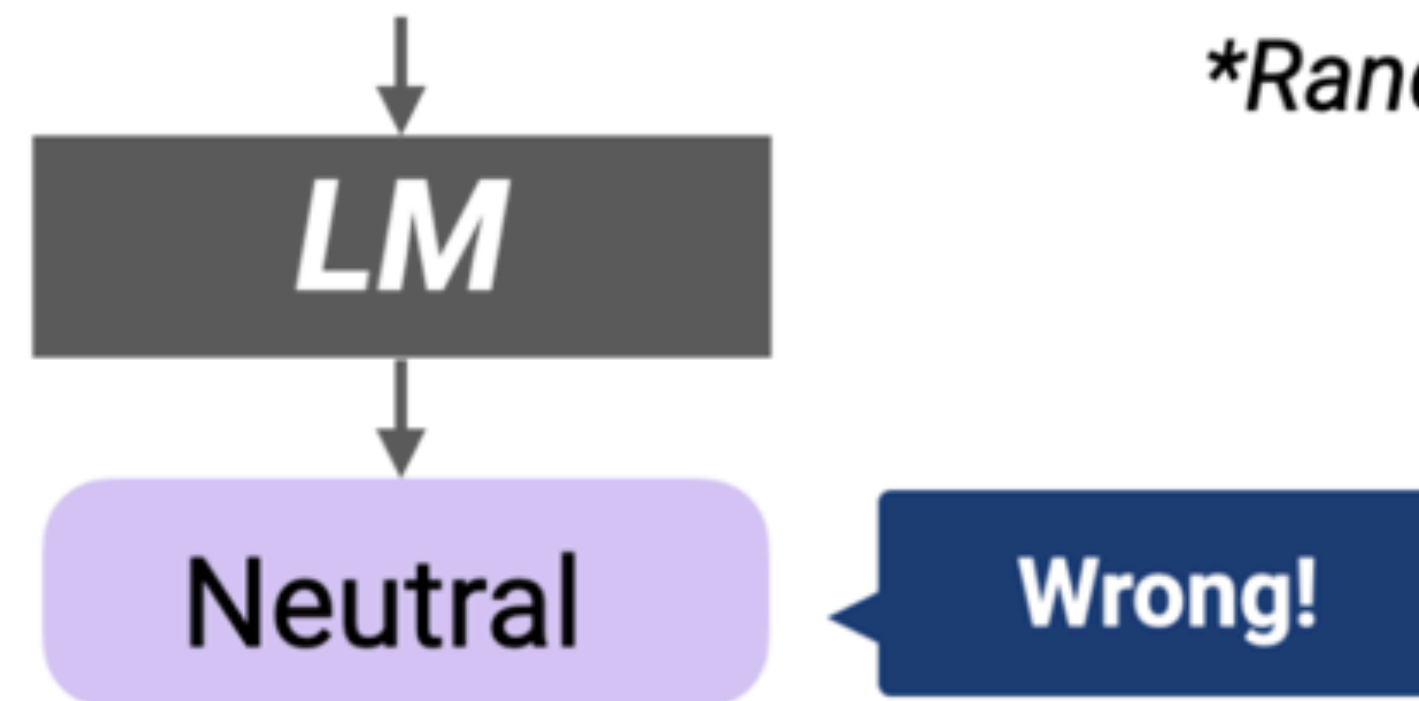
**Randomly Sampled from CC News*



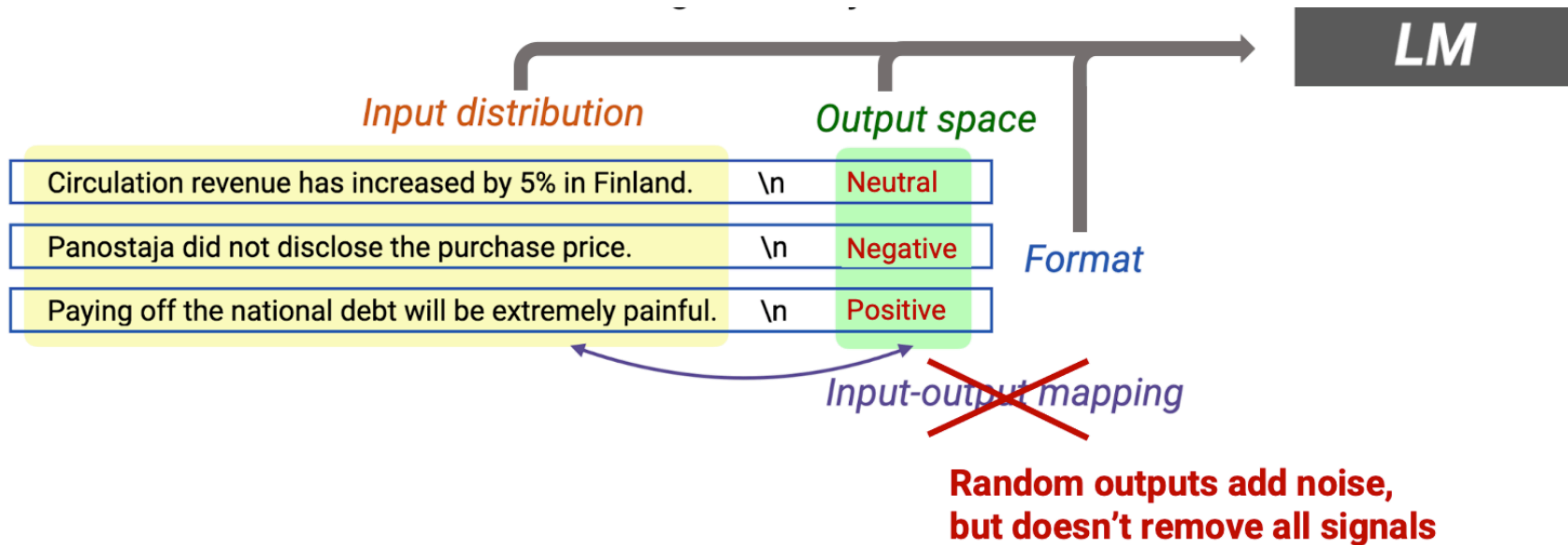
The input distribution matters: using inputs from an out of domain corpus causes a large performance drop

Circulation revenue has increased by 5% in Finland. \n Unanimity
Panostaja did not disclose the purchase price. \n Wave
Paying off the national debt will be extremely painful. \n Guana
The company anticipated its operating profit to improve. \n _____

**Random English unigrams*



The output distribution matters: using labels that are random English unigrams causes a large performance drop



Training examples (truncated)

```
beet: sport  
golf: animal  
horse: plant/vegetable  
corn: sport  
football: animal
```



Test input and predictions

```
monkey: plant/vegetable ✓  
panda: plant/vegetable ✓  
cucumber: sport ✓  
peas: sport ✓  
baseball: animal ✓  
tennis: animal ✓
```

An example synthetic task with unusual semantics that GPT-3 can successfully learn. A modified figure from Rong.

IN-CONTEXT LEARNING LEARNS LABEL RELATIONSHIPS BUT IS NOT CONVENTIONAL LEARNING

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In-Context Learning (ICL)

- How does the conditional label distribution of ICL examples affect accuracy?
- ICL does incorporate in-context label information and can even learn truly novel tasks in-context.
- Analogies between ICL and conventional learning algorithms fall short in a variety of ways
 - Label relationships inferred from pre-training have a lasting effect that cannot be surmounted by in-context observations
 - Additional prompting can improve but likely not overcome this deficiency
 - ICL does not treat all information provided in-context equally and preferentially makes use of label information that appears closer to the query

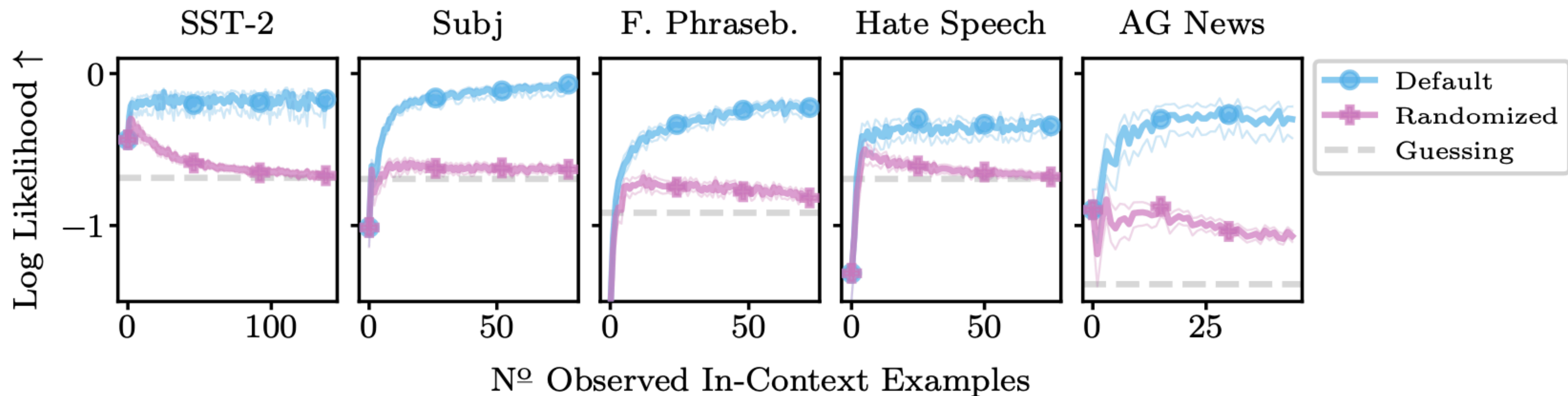


Figure 1: ICL predictions generally depend on the conditional label distribution of in-context examples: when in-context labels are **randomized**, average log likelihoods of label predictions decrease compared to ICL with **default** labels for LLaMa-2-70B across a variety of tasks. Results averaged over 500 in-context datasets and thin lines are 99 % confidence intervals. See §5 for details.

Table 1: Average differences between ICL log likelihoods for default and randomized labels. Bold entries indicate differences are statistically significant. We can disregard lightgray entries: for them, default ICL performance is not significantly better than a random guessing baseline. Whenever default ICL outperforms the baseline, ICL almost always performs significantly worse (positive differences) for random labels. Averages over 500 runs at max. context size, standard errors in Table F.1.

Δ Log Likelihood	SST-2	Subj	FP	HS	AGN	MQP	MRPC	RTE	WNLI
LLaMa-2 7B	0.42	0.39	0.57	0.18	0.53	0.03	0.02	0.03	0.02
LLaMa-2 13B	0.41	0.62	0.49	0.24	0.81	0.04	0.01	0.06	0.02
LLaMa-2 70B	0.51	0.53	0.57	0.34	0.80	0.29	0.04	0.22	0.18
Falcon 7B	0.20	0.19	0.25	0.06	0.31	0.01	0.01	−0.01	0.01
Falcon 7B Instr.	0.13	0.08	0.11	0.03	0.15	0.03	0.02	−0.00	0.00
Falcon 40B	0.34	0.35	0.31	0.18	0.90	0.06	0.01	0.01	0.02
Falcon 40B Instr.	0.25	0.37	0.27	0.02	0.77	0.06	0.02	0.02	0.04

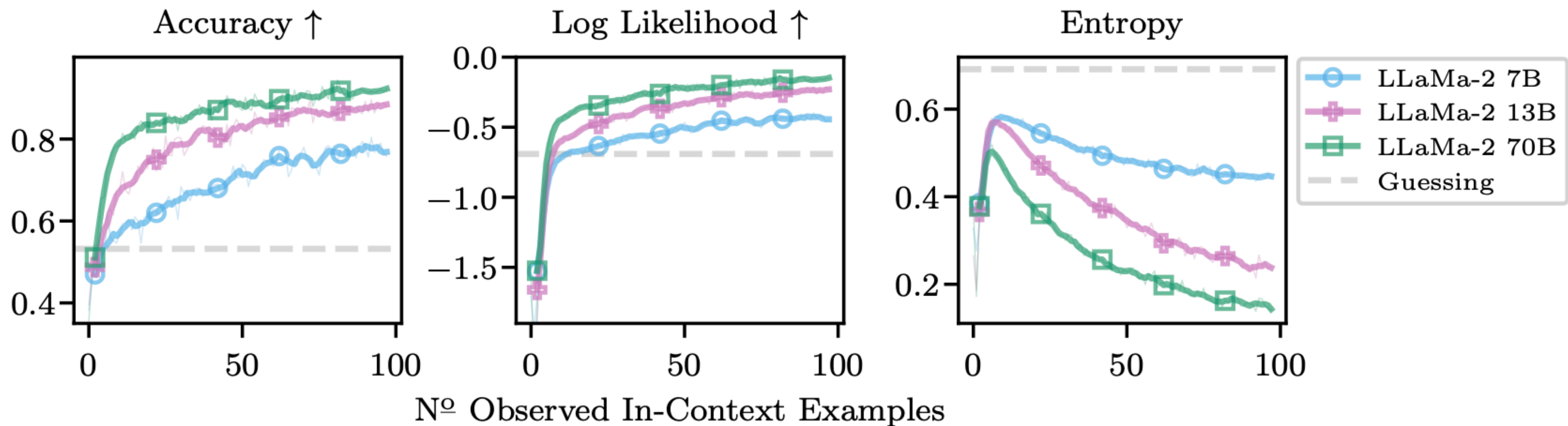


Figure 4: Few-shot ICL achieves accuracies significantly better than random guessing on our **novel author identification** task. Thus, LLMs can learn novel label relationships entirely in-context. Averages over 500 runs, thick lines with additional moving average (window size 5) for clarity.

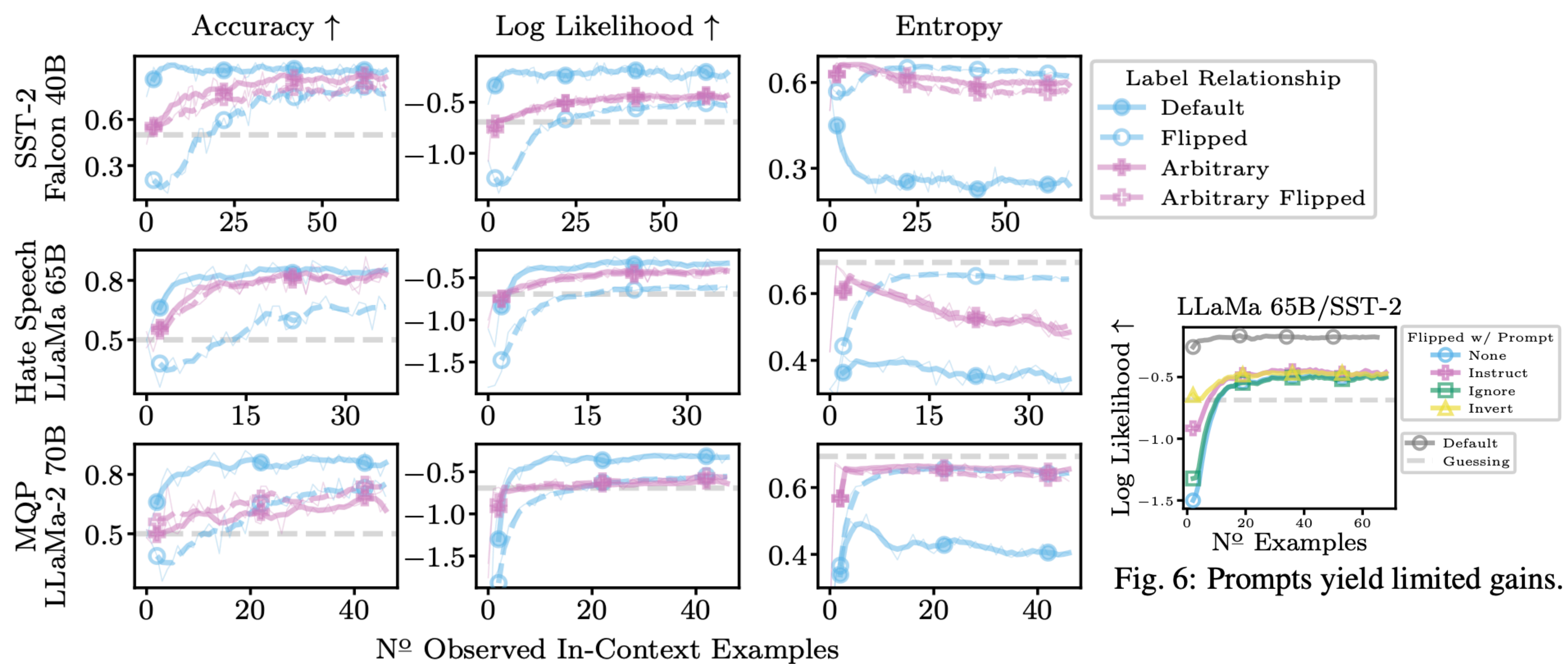


Fig. 6: Prompts yield limited gains.

Figure 5: Few-shot ICL with **replacement labels** for Falcon-40B on SST-2, LLaMa-2-65B on Hate Speech, and LLaMa-2-70B on MQP. Table 2 and §F contain results for all other models and tasks. ICL achieves better than guessing performance for all label relations and models. However, predictions for flipped labels (dashed blue) plateau at a higher entropies and lower likelihoods than those for the default label relation (solid blue). For arbitrary labels (pink), the model performs similarly for both label directions. Averages over 100 runs and thick lines with moving average (window size 5).

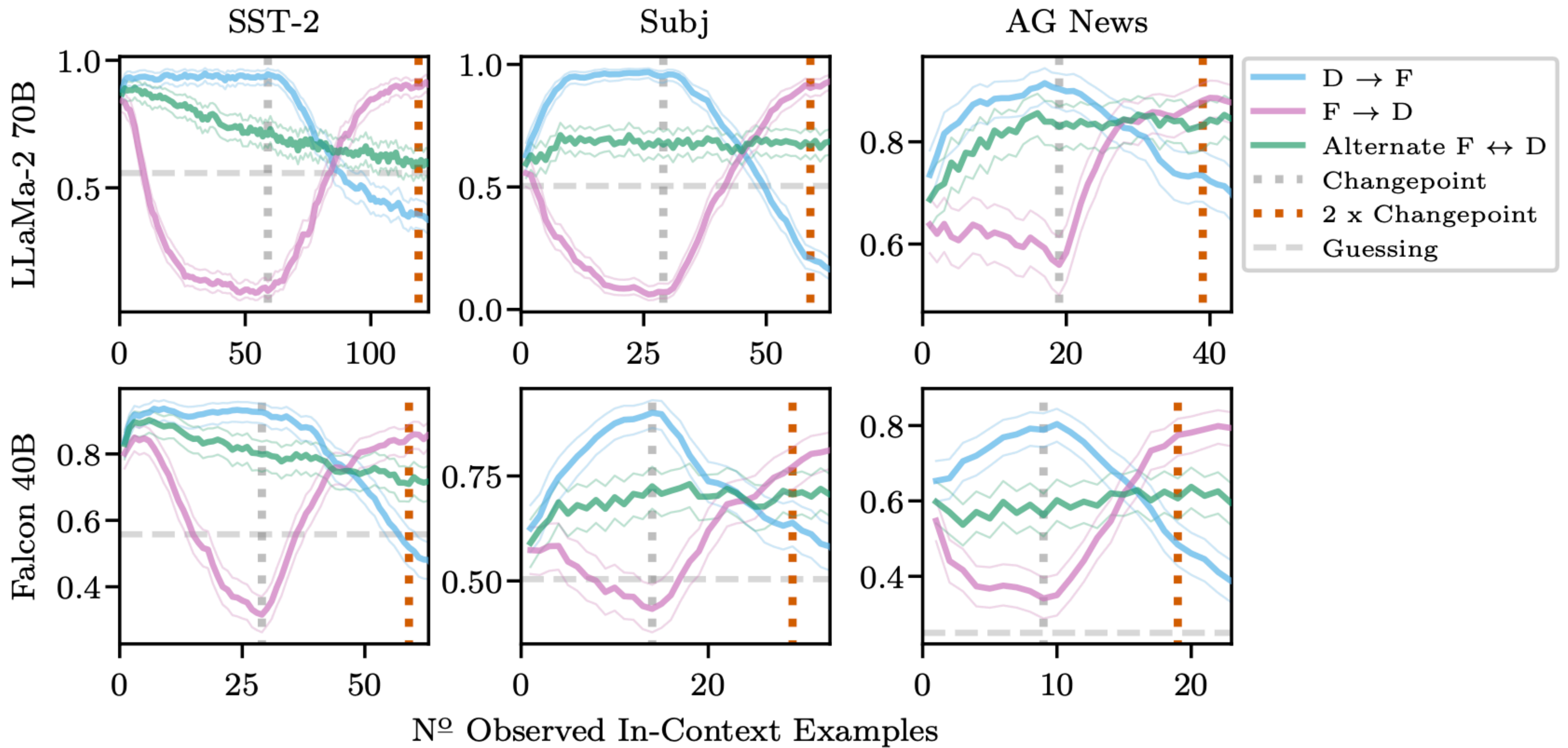
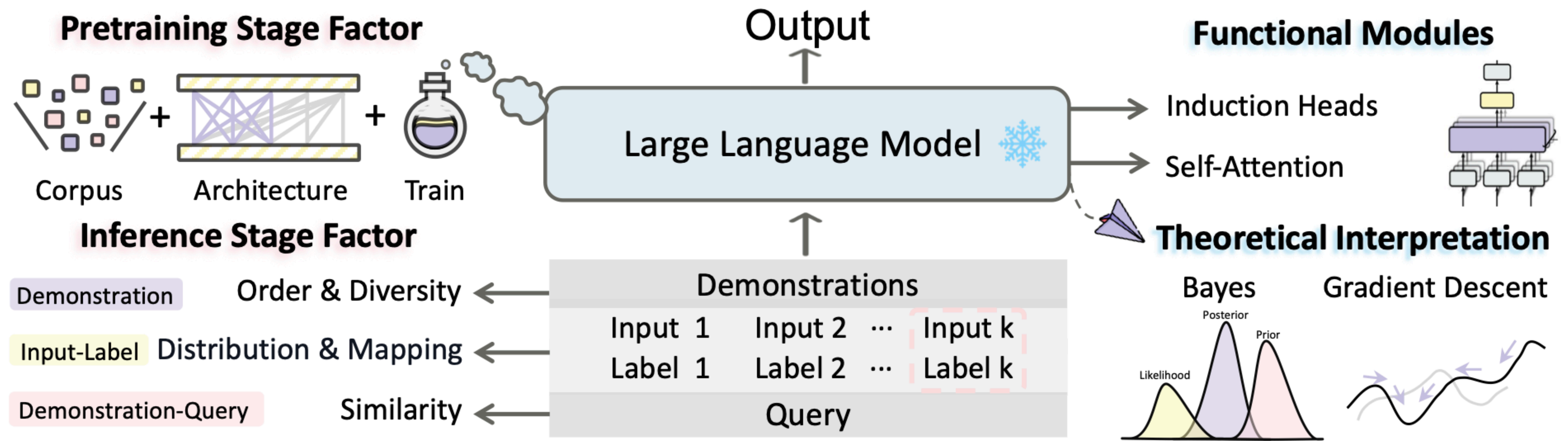


Figure 7: Few-shot ICL accuracies when the **label relationship changes throughout ICL**. For ($D \rightarrow F$), we start with default labels and change to flipped labels at the changepoint, for ($F \rightarrow D$) we change from flipped to the default labels at the changepoint, and for (Alternating $F \leftrightarrow D$) we alternate between the two label relationships after every observation. For all setups, at ‘2 x Changepoint’, the LLMs have observed the same number of examples for both label relations. If, according to NH3, ICL treats all in-context information equally, predictions should be equal at that point—but they are not. Bootstrapped 99 % confidence intervals, moving averages (size 3), and 500 repetitions.

Survey of papers about ICL <https://arxiv.org/abs/2301.00234>



Metaprompting

A prompt to help write prompts

Today you will be writing instructions to an eager, helpful, but inexperienced and unworldly AI assistant who needs careful instruction and examples to understand how best to behave. I will explain a task to you. You will write instructions that will direct the assistant on how best to accomplish the task consistently, accurately, and correctly. Here are some examples of tasks and instructions.

<Task Instruction Example>

<Task>

Answer questions about a document and provide references

</Task>

<Inputs>

{DOCUMENT}

{QUESTION}

</Inputs>

<Instructions>

I'm going to give you a document. Then I'm going to ask you a question about it. I'd like you to first write down exact quotes of parts of the document that would help answer the question, and then I'd like you to answer the question using facts from the quoted content. Here is the document:


```
<document>
{$DOCUMENT}
</document>
```

Here is the question:

```
<question>{$QUESTION}</question>
```

First, find the quotes from the document that are most relevant to answering the question, and then print them in numbered order. Quotes should be relatively short.

If there are no relevant quotes, write "No relevant quotes" instead.

Then, answer the question, starting with "Answer:". Do not include or reference quoted content verbatim in the answer. Don't say "According to Quote [1]" when answering. Instead make references to quotes relevant to each section of the answer solely by adding their bracketed numbers at the end of relevant sentences.

Thus, the format of your overall response should look like what's shown between the <example> tags. Make sure to follow the formatting and spacing exactly.


```
<example>
<Relevant Quotes>
<Quote> [1] "Company X reported revenue of $12 million in 2021." </Quote>
<Quote> [2] "Almost 90% of revenue came from widget sales, with gadget sales making up the remaining
10%." </Quote>
</Relevant Quotes>
<Answer>
[1] Company X earned $12 million. [2] Almost 90% of it was from widget sales.
</Answer>
</example>
```

If the question cannot be answered by the document, say so.

Answer the question immediately without preamble.

```
</Instructions>
```

```
</Task Instruction Example>
```

Final part of the metaprompt after several such examples is to format user input prompts into the same style as the examples.

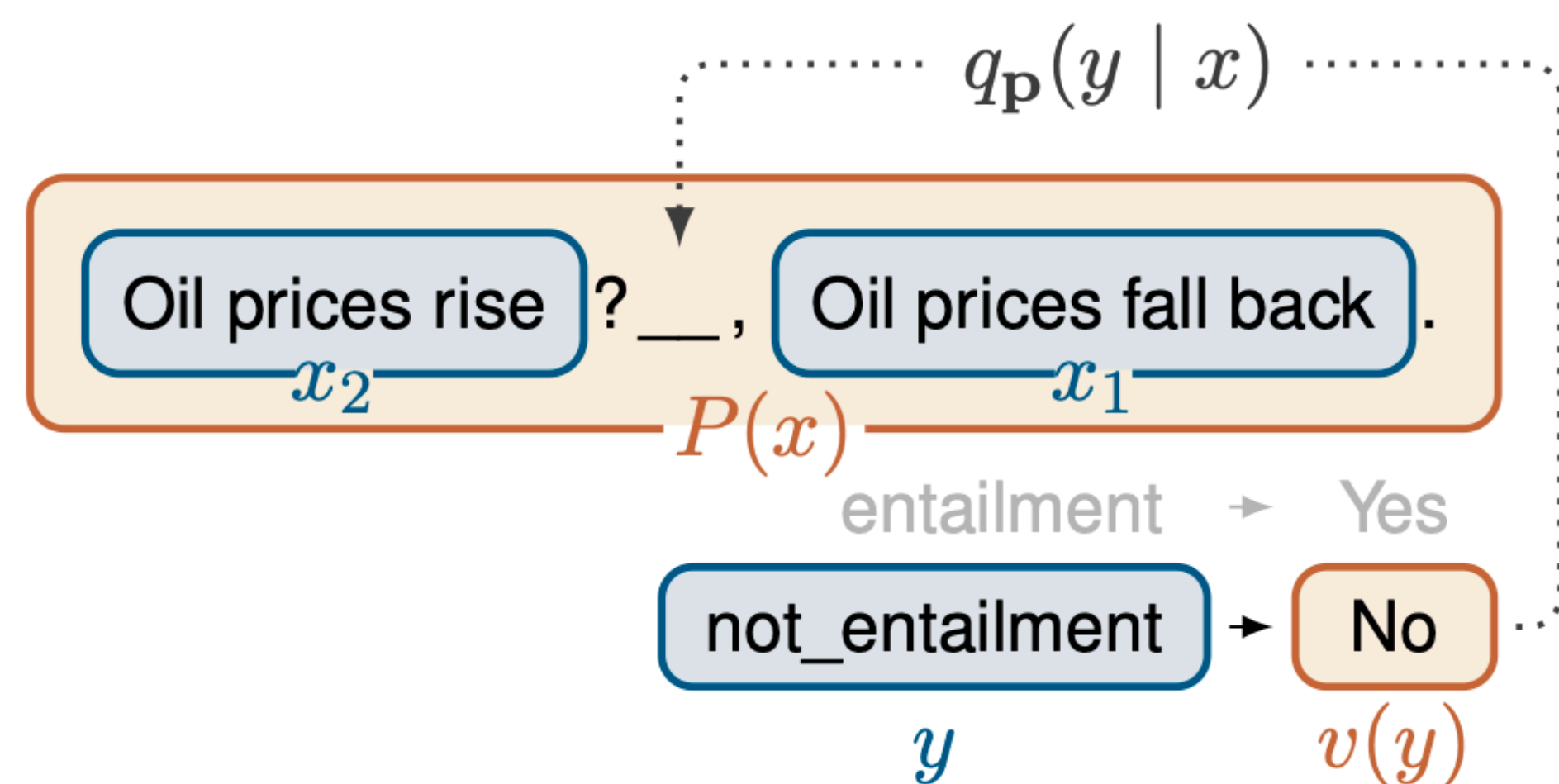
An excellent example of in-context learning.

[anthropic notebook on Colab](#)

Efficient few-shot learning

Prompt tuning: few-shot with smaller LMs

iPet: better pre-training for each task improves accuracy for small LMs



test	GPT-3	175,000	71.8	prompt
	PET	223	74.0	prompt FT
	iPET	223	75.4	prompt FT
	SotA	11,000	89.3	full FT

Figure 2: Application of a PVP $\mathbf{p} = (P, v)$ for recognizing textual entailment: An input $x = (x_1, x_2)$ is converted into a cloze question $P(x)$; $q_{\mathbf{p}}(y | x)$ for each y is derived from the probability of $v(y)$ being a plausible choice for the masked position.